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MATH 245A

Homework 1
Due: October 12, 2026

Problem 1.1

Let μ be an outer measure on a set X and let $\{A_k\}_{k=1}^{\infty}$ be a sequence of subsets of X.
(a) Prove that if $\mu[A_1] = 0$, then A_1 is μ-measurable.
(b) Prove that A_1 is μ-measurable if and only if its complement A_1^c is μ-measurable.
(c) Assume in the sequel that $\{A_k\}_{k=1}^{\infty}$ is a sequence of μ-measurable sets. Prove that if $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_k \subseteq \cdots$, then $\mu[A_1] + \mu[A_2 \setminus A_1] + \cdots + \mu[A_k \setminus A_{k-1}] = \mu[A_k]$.
(d) Prove that if $A_1 \supseteq A_2 \supseteq \cdots \supseteq A_k \supseteq \cdots$ and $\mu[A_1] < +\infty$ then $\lim_{k\to+\infty} \mu[A_k] = \mu[\cap_{k=1}^{\infty} A_k]$. Give a counterexample to this statement when $\mu[A_1] = +\infty$.
(e) Assume that $B \subseteq X$. Show that if A_1 is μ-measurable then A_1 is $\mu_{|B}$-measurable.

Solution. **(a)**: Suppose $\mu[A_1] = 0$. To show that A_1 is μ −measurable, it suffices to show that $\mu[B] \geq \mu[B - A_1] + \mu[A_1 \cap B]$ for all $B \subset X$. Fix $B \subseteq X$. Note that $B - A_1 \subseteq B$ and $A_1 \cap B \subseteq A$ hence $\mu[B - A_1] \leq \mu[B]$ and $\mu[A \cap B] = 0$. Hence $\mu[B] \geq \mu[B - A_1] = \mu[B - A_1] + \mu[A \cap B]$.

(b): Suppose A is μ −measurable, thus $\mu[B] = \mu[B - A] + \mu[A \cap B]$. We want to show that $\mu[B] \geq \mu[B - A^c] + \mu[A^c \cap B]$ for all $B \subseteq X$.

$$\begin{aligned} B - A^c &= (B \cap (A^c)^c) = B \cap A \\ B \cap A^c &= B - A \\ \Leftrightarrow \mu[B - A^c] + \mu[A^c \cap B] &= \mu[B \cap A] + \mu[B - A] \\ &= \mu[B] \end{aligned}$$

All implications are bidirectional.

(c): Suppose $\{A_k\}_{k=1}^{\infty}$ is a sequence of μ-measurable and $A_i \subseteq A_{i+1}$. Base case for $\mu[A_1]$ is trivial. Assume for the sake of induction that:

$$\sum_{k=1}^{n-1} \mu[A_{k+1} - A_k] + \mu[A_1] = \mu[A_n].$$

Now by μ −measurablity of A_n, we know

$$\begin{aligned} \mu[A_{n+1}] &= \mu[A_{n+1} - A_n] + \mu[A_{n+1} \cap A_n] \\ &= \mu[A_{n+1} - A_n] + \mu[A_n] \quad (A_n \subseteq A_{n+1}) \\ &= \sum_{k=1}^{n} \mu[A_{k+1} - A_k] + \mu[A_1] \quad (\text{inductive hyp.}) \end{aligned}$$

(d): Suppose

$$A_i \supseteq A_{i+1},$$

$\mu[A_1] < \infty$, and all A_i are μ-measurable. A_i^c is measurable.

Problem 1.2*

Suppose μ is a measure on $\mathbb{Z}^+ := \mathbb{Z} \cap [0, +\infty)$ such that every subset of $\mathbb{Z}^+$ is μ-measurable. Show that there exists a sequence $(\omega_k)_{k=0}^{\infty} \subseteq [0, +\infty]$ such that for every $E \subseteq \mathbb{Z}^+$ we have

$$\mu(E) = \sum_{k \in E} \omega_k.$$

Solution. Let

$$\omega_k := \mu[\{k\}].$$

For finite sets $E \subset \mathbb{Z}^+$, showing the claim follows from a inductive argument. The case for singletons is trivial. Suppose for the sake of induction that it is true for all $E \subset \mathbb{Z}^+$ such that $|E| = k$. Then let $F = \{x_1, ..., x_{k+1}\} \subset \mathbb{Z}^+$.

$$\begin{aligned} \mu[F] &= \mu[F - \{x_{k+1}\}] + \mu[\{x_{k+1}\}] \quad (F \text{ is measurable}) \\ &= \sum_{i=1}^{k} \omega_{x_i} + \omega_{x_{k+1}}. \end{aligned}$$

Now for $E \subseteq \mathbb{Z}^+$ infinite we will do the following argument. E countable, thus let $E = \{x_i\}_{i=1}^{\infty}$ be an enumeration. Let $A_n := \{x_1, ..., x_n\}$. Note that $A_n \subseteq A_{n+1}$.

$$\begin{aligned} \mu[E] &= \lim_{n \to \infty} \mu[A_n] \\ &= \lim_{n \to \infty} \left(\mu[A_1] + \sum_{k=1}^{n-1} \mu[A_{k+1} - A_k] \right) \\ &= \lim_{n \to \infty} \left(\omega_{x_1} + \sum_{k=1}^{n-1} \omega_{x_k} \right) \\ &= \sum_{k=1}^{\infty} \omega_{x_k} = \sum_{e \in E} \omega_e \end{aligned}$$

Problem 1.3*

Suppose μ is an outer measure on X such that $\mu(X) < \infty$ and suppose S is the set of μ-measurable subsets. Suppose $\mathcal{A} \subseteq S$ is such that for every $A \in \mathcal{A}$, $\mu(A) > 0$ and for every $A, B \in \mathcal{A}$ such that $A \neq B$ then $A \cap B = \emptyset$. Show that $\mathcal{A}$ is countable.

Solution. Let

$$I_n := \left\{ A \in \mathcal{A} : \mu[A] \geq \frac{1}{n} \right\}.$$

Note that

$$\mathcal{A} = \bigcup_{n \in \mathbb{N}} I_n.$$

Assume the contrary that I_n is not finite. Let $J \subseteq I_n$ be a countably infinite subset,

$$\mu[X] \geq \mu\left[\bigcup_{A \in I_n} A\right] \geq \mu\left[\bigcup_{A \in J} A\right] = \sum_{A \in J} \mu[A] \geq \sum_{A \in J} \frac{1}{n} = \infty.$$

Thus a contradiction occurs. Hence $|I_n|$ is finite. Hence $\mathcal{A}$ is a countable union of finite sets hence must be countable.

Problem 1.4*

Let μ be an outer measure on a set X and let S be the collection of μ-measurable subsets on X. Show that we cannot have

$$\{\mu(E) : E \in S\} = [0, 1).$$

Solution. Note that $X \in S$. Thus $\exists \alpha \in [0, 1)$ where $\mu[X] = \alpha$. But since we have a bijection between $\{\mu(E) : E \in S\} = [0, 1)$ there exists a $E \in S$ such that $\mu(S) = \frac{\alpha+1}{2}$. But $E \subseteq X$ and by monotonicity of outer measure, $\frac{\alpha+1}{2} = \mu(E) \leq \mu(X) = \alpha$ this is a contradiction.

Problem 1.5*

Let μ be a measure on a set X such that $\mu[X] = 1$ and let $\mathcal{M}$ be the set of μ-measurable sets. Suppose that $\mu[M] > 0$ for each $\emptyset \neq M \in \mathcal{M}$ and let

$$\alpha(x) = \inf_{M \in \mathcal{M}} \{\mu[M] : x \in M\} \quad (x \in X).$$

(a) Show that there exists a set $A_x \in \mathcal{M}$ such that $x \in A_x$ and $\mu[A_x] = \alpha(x)$.
(b) Show that the sets $\{A_x\}$ are either disjoint or identical.

Solution. **(a)**: We know that following that: $\forall n \in \mathbb{N}$ there exists a $E_n \in \mathcal{M}$ such that

$$\mu(E_n) \leq \alpha(x) + \frac{1}{n}$$

and $x \in E_n$.

$$\bigcap_{n \in \mathbb{N}} E_n \ni \{x\}.$$

Consider

$$F_k := \bigcap_{n=1}^{k} E_k.$$

Note that $F_1 \supseteq F_2 \supseteq ...$, and $\{x\} \in F_k \Rightarrow \alpha(x) \leq \mu(F_n) \leq \mu(E_n) = \alpha(x) + \frac{1}{n}$. By problem 1, we know that

$$\lim_{n \to \infty} \mu[F_n] = \mu\left[\bigcap_{n \in \mathbb{N}} F_n\right].$$

By the squeeze theorem we know, that

$$\alpha(x) \leq \mu\left[\bigcap_{n\in\mathbb{N}} F_n\right] \leq \lim_{n\to\infty}\left(\alpha(x) + \frac{1}{n}\right) = \alpha(x) \Rightarrow \mu\left[\bigcap_{n\in\mathbb{N}} F_n\right] = \alpha(x).$$

Furthermore, we know that

$$\bigcap_{n\in\mathbb{N}} F_n = \bigcap_{n\in\mathbb{N}} E_n \text{ is measurable..}$$

Thus

$$A_x = \bigcap_{n\in\mathbb{N}} F_n$$

(b): Suppose $x, y \in X$ where $A_x \neq A_y$. Assume the contrary, that $A_x \cap A_y \neq \emptyset$.

First note that if $M \in \mathcal{M}$ with $x \in M$, then $A_x \subseteq M$. Indeed, $A_x \cap M \in \mathcal{M}$ and $x \in A_x \cap M$, so

$$\alpha(x) \leq \mu[A_x \cap M] \leq \mu[A_x] = \alpha(x).$$

Since $\mu[X] = 1 < \infty$, $\mu[A_x - M] = \mu[A_x] - \mu[A_x \cap M] = 0$. But $A_x - M \in \mathcal{M}$, so $A_x - M = \emptyset$ by assumption. Thus $A_x \subseteq M$.

Take $z \in A_x \cap A_y$. Since $A_x \in \mathcal{M}$ and $z \in A_x$, we have $A_z \subseteq A_x$, and similarly $A_z \subseteq A_y$.

Case 1: $x \in A_z$. Then $A_x \subseteq A_z$ by the above, so $A_x = A_z$.

Case 2: $x \notin A_z$. Then $x \in A_x - A_z \in \mathcal{M}$, so $A_x \subseteq A_x - A_z$. But $z \in A_x$ and $z \in A_z$, so $z \notin A_x - A_z$, a contradiction. Hence Case 2 cannot occur.

So $A_x = A_z$, and by the same argument applied to y, $A_y = A_z$. Thus $A_x = A_y$, a contradiction. Hence $A_x \cap A_y = \emptyset$.

Problem 1.6

Let μ be a measure on X such that $\mu[X] < \infty$. Show that if $\{x\}$ is μ-measurable for every $x \in X$ then there are at most countably many $x \in X$ such that $\mu[\{x\}] > 0$.

Solution. Let $Y \coloneqq \{x \in X : \mu(\{x\}) > 0\}$. For all $n \in \mathbb{N}$, let $Y_n \coloneqq \left\{y \in Y : \mu(\{y\}) \geq \frac{1}{n}\right\}$. We will show that $|Y_n| < \infty$. Assume the contrary, that Y_n is not finite. Let $\{y_k\}_{k=1}^{\infty} \subseteq Y$ be a enumeration of countably infinite elements of Y. By monotonicity:

$$\mu(X) \geq \mu(Y) \geq \mu(\{y_k : k \in \mathbb{N}\}) = \sum_{k=1}^{\infty} \mu(\{y_k\}) \geq \sum_{k=1}^{\infty} \frac{1}{n} = \infty.$$

Thus $\mu(X)$ is not finite and a contradiction occurs. Hence $|Y_n|$ must be finite. Y is the countable union of finite sets (with possible intersections), hence must be countable.

Problem 1.7*

Show that $S = \{\cup_{n\in K}(n, n+1] : K \subseteq \mathbb{Z}\} \cup \{\emptyset\}$ is a σ-algebra on $\mathbb{R}$.

Solution. For $S \subset \mathbb{Z}$, let

$$U_K := \bigcup_{n\in S}(n, n+1].$$

We have $\emptyset \in S$ and

$$\mathbb{R} = \bigcup_{n\in\mathbb{Z}}(n, n+1] \in S.$$

Now suppose $K \subset \mathbb{Z}$ where $K \neq \emptyset$. Consider $\mathbb{Z} \cap K^c$ and the set associated with it in S,

$$U_K \cap U_{\mathbb{Z}\cap K^c} = \bigcup_{(n,m)\in K\times(\mathbb{Z}\cap K^c)} (m, m+1] \cap (n, n+1].$$

Note that $(n, n+1] \cap (m, m+1] \neq \emptyset \Leftrightarrow n = m$, but $K \cap (\mathbb{Z} \times K^c) = \emptyset$ which means that

$$U_K \cap U_{\mathbb{Z}\cap K^c} = \emptyset.$$

For all $r \in \mathbb{R}$, let $k = \lfloor r \rfloor$. If $k \in \mathbb{Z}$, $r \in K \cup (\mathbb{Z} \cap K^c)$. Cases:
1. $k \in K$ and $k-1 \in K$: Thus $r \in (k, k+1] \cup (k-1, k] \subset U_K$.
2. $k \in K$ and $k-1 \in \mathbb{Z} \cap K^c$: Thus $r \in (k, k+1] \subset U_K$ or $r \in (k-1, k] \subseteq U_{\mathbb{Z}\cap K^c}$. Thus $r \in U_K \cap U_{\mathbb{Z}\cap K^c}$

The other two cases are symmetric to the first two. Hence $r \in U_k \cup U_{\mathbb{Z}\cap K^c}$. Hence $U_K^c = U_{\mathbb{Z}\cap K^c} \in S$. Thus S is closed under complements. Note for any $U \in S$, $\exists K \subseteq Z$ such that $U = U_K$, by definition. Let $\{K_i\}_{i=1}^{\infty}$ where $K_i \subseteq \mathbb{Z}$. Let

$$K = \bigcup_{i=1}^{\infty} K_i \subseteq Z$$

Note the following,

$$\begin{aligned}\bigcup_{i=1}^{\infty} U_{K_i} &= \bigcup_{i=1}^{\infty}\bigcup_{n\in K_i}(n, n+1] \\ &= \bigcup_{\substack{n\in\mathbb{Z} \\ \exists i\in\mathbb{N}, n\in K_i}} (n, n+1] \\ &= \bigcup_{n\in K}(n, n+1] = U_K \in S.\end{aligned}$$

Thus S is closed under countable union, hence S is a σ-algebra.

For the sake of brevity, let $\mathcal{T}(X)$ denote the topology of a topological space X.

Problem 1.8*

Prove that $\sigma(C_1) = \sigma(C_2) = \sigma(C_3) = \mathcal{B}$, where $\mathcal{B}$ is the Borel σ-algebra on $\mathbb{R}$. Here,

$$C_1 = \{(r, s] : r, s \in \mathbb{Q}\}, \quad C_2 = \{(r, n] : r \in \mathbb{Q}, n \in \mathbb{Z}\}, \quad C_3 = \{[r, +\infty) : r \in \mathbb{Q}\}.$$

Solution. To show $\sigma(C_i) \supset \mathcal{B}$, we will show for all $U \in \mathcal{T}(\mathbb{R})$, $U \in \sigma(C_i)$ and since $\mathcal{B}$ is the smallest σ-algebra containing $\mathcal{T}(\mathbb{R})$. Hence if $\mathcal{T}(\mathbb{R}) \subseteq \sigma(C_i) \Rightarrow \mathcal{B} \subseteq \sigma(C_i)$. Moreover it suffices to show that $(a,b) \in \sigma(C_i)$ for any $a, b \in \mathbb{R}$ $(a < b)$ because all open sets are countable unions of open intervals (metric space plus separability).Fix a $a, b \in \mathbb{R}$ where $a < b$. Let $(r_n), (s_n)$ be sequences in $\mathbb{Q}$ such that r_n is monotone nonincreasing where $r_n \to a$, and s_n monotone nondecreasing such that $s_n \to b$ and $s_n \neq b$. Let $A_n \coloneqq (r_n, s_n]$ and note that $A_n \subseteq A_{n+1} \subseteq (a,b)$. Thus

$$A = \bigcup_{n=1}^{\infty} (r_n, s_n] \subseteq (a,b).$$

We need to show the opposite containment is true as well. Take $x \in (a,b)$. There exists a $N \in \mathbb{N}$ such that $\forall n \leq N$ we have that

$$a \leq r_n < x.$$

Similarly, $\exists M \in \mathbb{N}$ such that for all $m \in M$ we have that

$$x < s_m < b.$$

Take $K = \max(N, M)$ and thus $r_K < x < s_K \Rightarrow x \in (r_k, s_k)$ and $x \in A$. Thus

$$(a,b) = A \in \sigma(C_1).$$

Thus

$$\mathcal{B} \subseteq \sigma(C_1).$$

Showing, $C_1 \subseteq \mathcal{B} \Rightarrow \sigma(C_1) \subseteq \mathcal{B}$. Fix a arbitrary $r, s \in \mathbb{Q}$ where $r < s$. Let (b_n) be a monotone nonincreasing sequence of real numbers such that $b_n \to s$ and $b_n \neq s$. Let $A_n \coloneqq (r, b_n)$. We will show that

$$\bigcap_{n \in \mathbb{N}} (r, b_n) = (r, s].$$

Note the following, $(r,s] \subseteq (r, b_n)$. Thus the nontrivial direction is showing that

$$A = \bigcap_{n \in \mathbb{N}} (r, b_n) \subseteq (r, s].$$

It suffices to show that $(s, \infty) \cap A = \emptyset$. Let $x \in (s, \infty)$. By assumption, there exists a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have that

$$s < b_n < x.$$

Thus $x \notin (r, b_n) \Rightarrow x \notin A$. Thus $A \cap (s, \infty) = \emptyset$. Thus $(r,s] = A \in \mathcal{B}$. Hence $\mathcal{B} = \sigma(C_1)$. Note some trivial directions, $C_2 \subseteq C_1 \Rightarrow \sigma(C_2) \subseteq \sigma(C_1)$.

1. $\sigma(C_2) = \sigma(C_1)$: To prove this we need to show that $\sigma(C_1) \subseteq \sigma(C_2)$, which implied by the fact that $C_1 \subseteq \sigma(C_2)$. For $q \in \mathbb{Q}$,

$$(q, \infty) = \bigcup_{n \in \mathbb{N}, n > q} (q, n] \in \sigma(C_2),$$

a countable union of sets in C_2. Now fix $r, s \in \mathbb{Q}$ with $r < s$. Then

$$(r, s] = (r, \infty) \cap (s, \infty)^c \in \sigma(C_2),$$

since $\sigma(C_2)$ is closed under complements and finite intersections.

Thus $C_1 \subseteq \sigma(C_2) \Rightarrow \sigma(C_1) \subseteq \sigma(C_2) \Rightarrow \sigma(C_1) = \sigma(C_2)$.

2. $\mathcal{B} = \sigma(C_3)$: To do this we will show that $C_3 \subseteq \mathcal{B}$ and $\mathcal{T}(\mathbb{R}) \subseteq \sigma(C_3)$. Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers such that $a_n \to r$ and a_n monotone increasing where $a_n \neq r$. Let $A_n = (a_n, \infty)$, and note $[r, \infty)$. Consider

$$A = \bigcap_{n \in \mathbb{N}} A_n \supseteq [r, \infty).$$

It suffices to show that,

$$A \cap (-\infty, r) = \emptyset.$$

For all $x \in (-\infty, r)$, $\exists N \in \mathbb{N}$ such that

$$x < a_n < r.$$

Thus $x \notin A_n \Rightarrow x \notin A$. Hence $[r, \infty) = A \in \mathcal{B}$. Thus $C_3 \subseteq \mathcal{B} \Rightarrow \sigma(C_3) \subseteq \mathcal{B}$.

For the reverse, it suffices to show $C_1 \subseteq \sigma(C_3)$, since then $\mathcal{B} = \sigma(C_1) \subseteq \sigma(C_3)$. For $q \in \mathbb{Q}$,

$$(q, \infty) = \bigcup_{n \in \mathbb{N}} \left[q + \frac{1}{n}, \infty\right) \in \sigma(C_3),$$

since each $q + \frac{1}{n} \in \mathbb{Q}$ and $\sigma(C_3)$ is closed under countable unions. Now fix $r, s \in \mathbb{Q}$ with $r < s$. Then

$$(r, s] = (r, \infty) \cap (s, \infty)^c \in \sigma(C_3),$$

as $\sigma(C_3)$ is closed under complements and finite intersections. Hence $C_1 \subseteq \sigma(C_3)$, so $\mathcal{B} \subseteq \sigma(C_3)$, and therefore $\sigma(C_3) = \mathcal{B}$.

Combining 1 and 2, $\sigma(C_1) = \sigma(C_2) = \sigma(C_3) = \mathcal{B}$. ■

Problem 1.9*

Prove that if $B \subseteq \mathbb{R}^d$ is a Borel set, $t \in \mathbb{R}$ and $c \in \mathbb{R}^d$ then $tB = \{tx : x \in B\}$ and $c + B = \{c + x : x \in B\}$ are Borel sets, where $\mathcal{B}$ is the Borel σ-algebra on $\mathbb{R}$.

Solution. For a function $f : X \to Y$. Let $f_* : 2^{2^X} \to 2^{2^Y}$ where

$$f_*(S) \coloneqq \{f(A) : A \subseteq S\}.$$

Fix a arbitrary $(c, t) \in \mathbb{R} \times \mathbb{R}^d$. $f_{c,t}(x) \coloneqq tx + c$ is a continuous function on $\mathbb{R}^d$. We have two cases, $t = 0$ and $t \neq 0$.

1. $t = 0$: Then $f_{c,0} = c$, meaning $f_{c,0}(X) = \{c\}$ for any $X \subseteq \mathbb{R}^d$ (not necessarily Borel).

$$\{c\} = \bigcap_{n\in\mathbb{N}} B\left(c, \frac{1}{n}\right) \Rightarrow \{c\} \in \mathcal{B}(\mathbb{R}^d).$$

2. $t \neq 0$: $f_{c,t}$ is a homemorphism, with its continuous inverse: $g_{c,t}(x) = \frac{1}{t}(x - c)$. This implies $\left(f_{c,t}\right)_*(\mathcal{T}(\mathbb{R})) \subseteq \mathcal{T}(\mathbb{R}) \Rightarrow \left(f_{c,t}\right)_*\left(\mathcal{B}(\mathbb{R}^d)\right) \subseteq \mathcal{B}(\mathbb{R}^d)$.

Thus $\left(f_{c,t}\right)_*\left(\mathcal{B}(\mathbb{R}^d)\right) \subseteq \mathcal{B}(\mathbb{R}^d)$ for any choice of $t \in \mathbb{R}$. Thus if $B \subseteq \mathcal{B}(\mathbb{R}^d)$, then $f_{c,1}(B) = c + B \in \mathcal{B}(\mathbb{R}^d)$ and $f_{0,t}(B) = tB \in \mathcal{B}(\mathbb{R}^d)$.

Problem 1.10

Let (X, μ) be an outer measure space and $(A_n)_n$ be a sequence of μ-measurable subsets of X. Show that if $\sum_{n=1}^{\infty} \mu[A_n] < \infty$ then $\mu[A] = 0$ if we set

$$A = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n.$$

Solution. Let

$$B_m = \bigcup_{n=m}^{\infty} A_m.$$

By monotonicity, $\mu[A] \leq \mu[B_m]$ for all $m \in \mathbb{N}$. Note that $B_m \supseteq B_{m+1}$. By subadditivity,

$$\mu[B_1] \leq \sum_{k=1}^{\infty} \mu[A_k] = L < \infty.$$

By Cauchy Criterion we know that for all $p \in \mathbb{N}$, there $\exists N_p \in \mathbb{N}$ such that if $N_p \leq n \leq m$ we have that

$$\left|\sum_{k=1}^{n} \mu[A_k] - \sum_{k=1}^{m} \mu[A_m]\right| < 2^{-p} \Rightarrow \sum_{k=n+1}^{m} \mu[A_k] < 2^{-p}.$$

Take $\lim_{m\to\infty}$ on both sides, and thus we see that for all $n \geq N_p$

$$\sum_{k=n+1}^{\infty} \mu[A_k] \leq 2^{-p}.$$

Note that

$$\mu[B_{n+1}] \leq \sum_{k=n+1}^{m} \mu[A_k] \leq 2^{-p}$$

for all $n \geq N_p$. $\forall \varepsilon > 0$, there exists a $P \in \mathbb{N}$ such that $2^{-P} < \varepsilon$. For all $n \geq N_P + 1$, we have that

$$\mu[B_n] \leq 2^{-P} < \varepsilon.$$

Hence $\lim_{n\to\infty} \mu[B_n] = 0$. By Problem 1c, we can conclude that

$$\mu[A] = \lim_{n\to\infty} \mu[B_n] = 0$$